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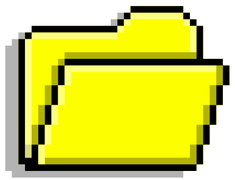
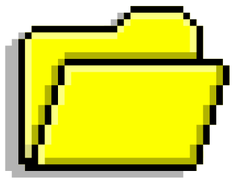
Presentation
Group 8
2026

Today's Agenda



(01) Task Description

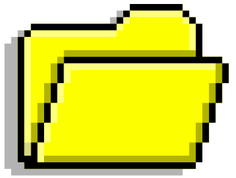


(02) Workload Distribution

(03) Block Diagrams

(04) Component Breakdown

(05) Project Management



(06) Short Demo

Task Description

Record and analyze audio samples by using two STM32s for sampling and processing the audio samples.

Key Features:

- Sampling
- Processing
- User Interface
- Output Formats

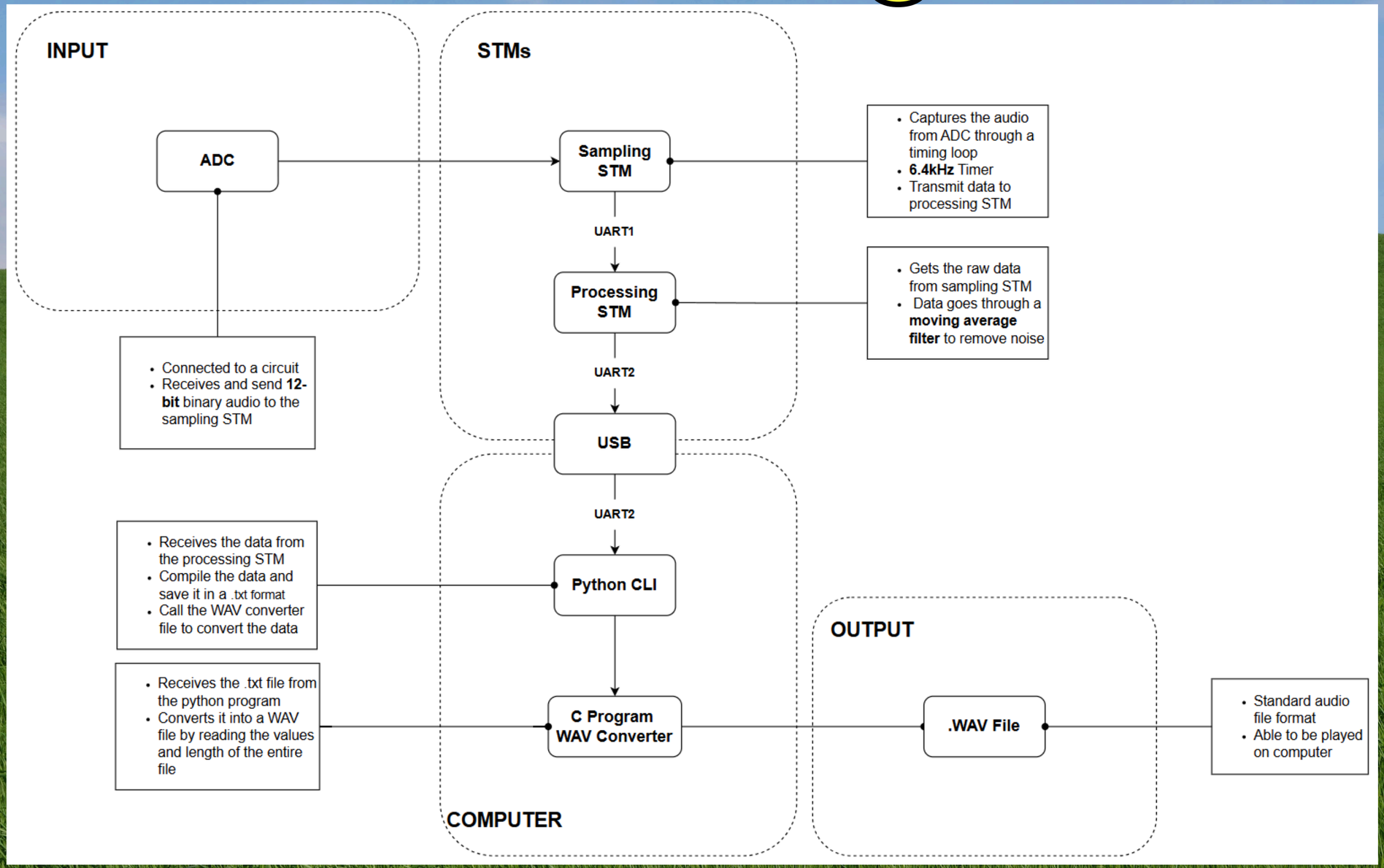
Workload Distribution

- Sampling STM - Man Kitt
- Processing STM - Hans
- Python CLI - Limin
- Project Management - Nicholas
- Presentation Slides - Rohan

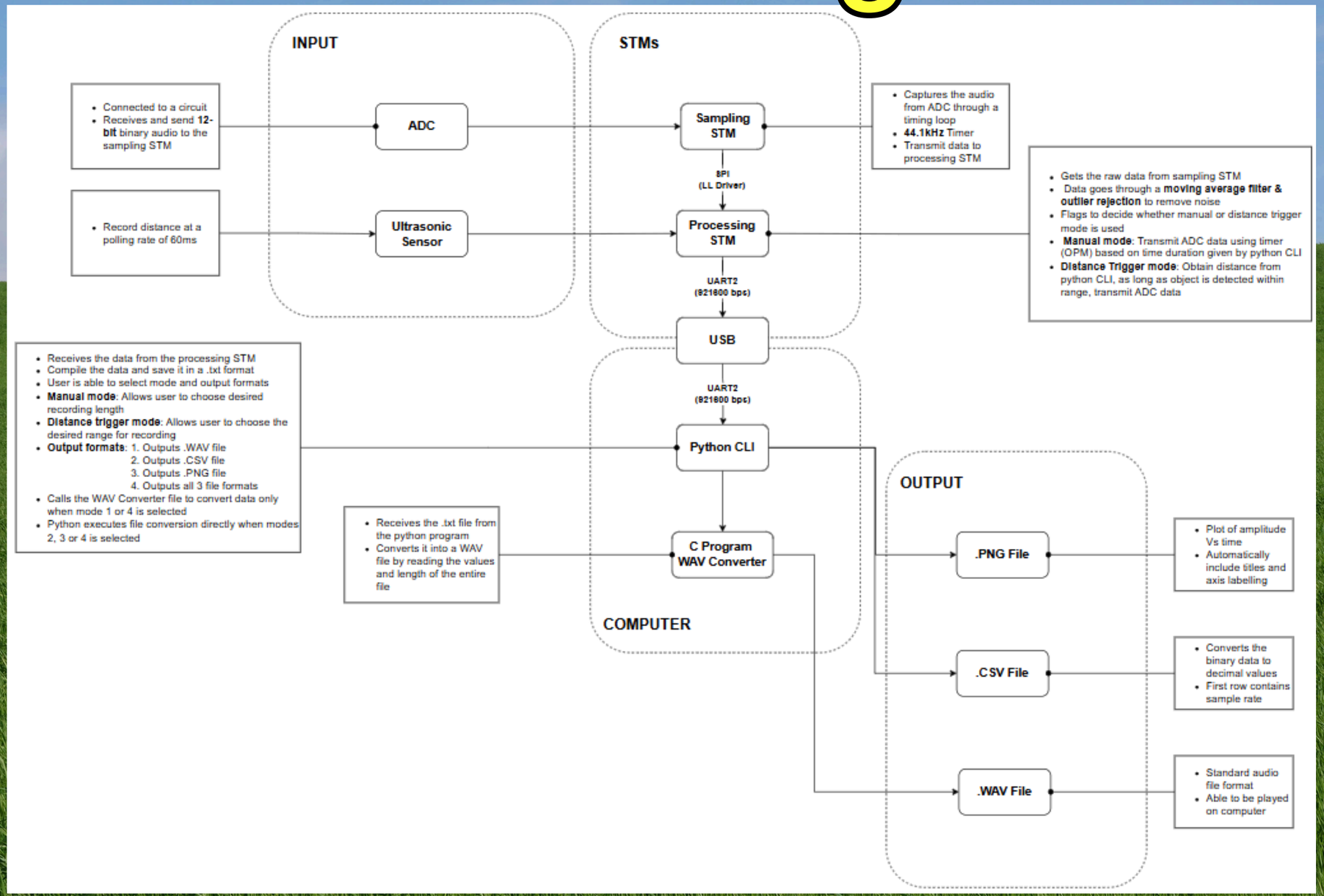
Block Diagrams

Shows the relation between each component and the flow of the system

Initial Design



Final Design

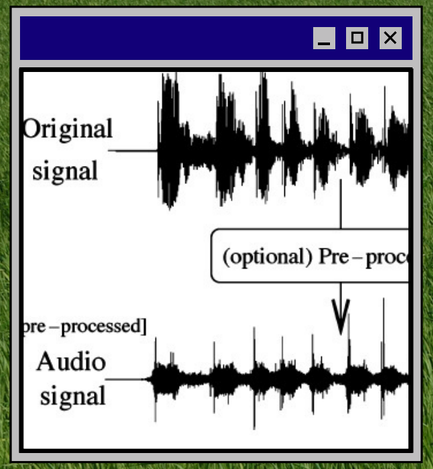


Component Breakdown

Brief description on how each component works

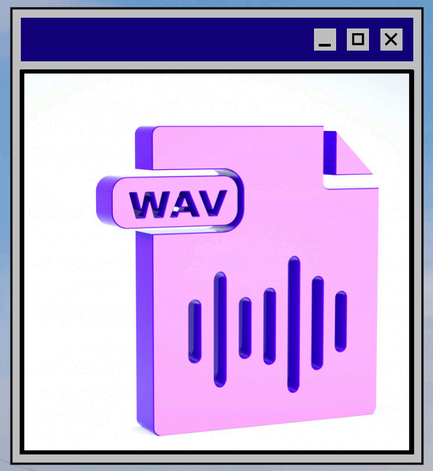
Sampling STM

- Audio Conversion: Continuously monitors an analog input via the ADC (Analog-to-Digital Converter), transforming sound waves into digital values at 44.1 kHz with 12-bit resolution.
- Direct Streaming: Samples are immediately forwarded upon conversion — each 12-bit value is packed into a single 16-bit word and transmitted to the Processing STM via SPI using the Low-Layer (LL) driver for minimal overhead.
- Data Packaging: Each 12-bit ADC result is placed into the lower 12 bits of a 16-bit SPI word and transmitted as a single atomic transaction, preserving sample integrity without splitting across multiple packets.
- Timing Accuracy: Uses TIM1's hardware TRGO trigger output to directly initiate ADC conversions at a fixed 44.1 kHz rate — no software interrupt is involved, ensuring deterministic and jitter-free sampling that keeps audio quality undistorted.



Processing STM

- **Proximity Detection:** Interfaces with ultrasonic sensor to detect when an object is within a specific range.
- **Event Triggering:** Signals itself to start or stop transmitting ADC data based on the proximity distance.
- **Signal Pre-processing:** Performs basic operations on the audio stream, such as average filtering and outlier removal.
- **Communication Hub:** Coordinates the flow of information between the ultrasonic sensor, the sampling STM, and the external user interface.
- **PC Communication:** UART @921600 bps that can support transmitting ADC data @44.1 kbps @12-bit




```
==========================================
  Main Menu
==========================================
    1. Recording Mode        (Enter: m)
    2. Distance Trigger Mode  (Enter: d)
    3. Exit                  (Enter: exit)
------------------------------------------
Choose mode: m
  Enter recording time (seconds): 10

------------------------------------------
Select output formats (use comma to separate):
  1. WAV
  2. CSV
  3. PNG
  4. ALL

------------------------------------------
  Enter: 4
No valid format selected, defaulting to all.
  Selected: CSV, PNG, WAV
```

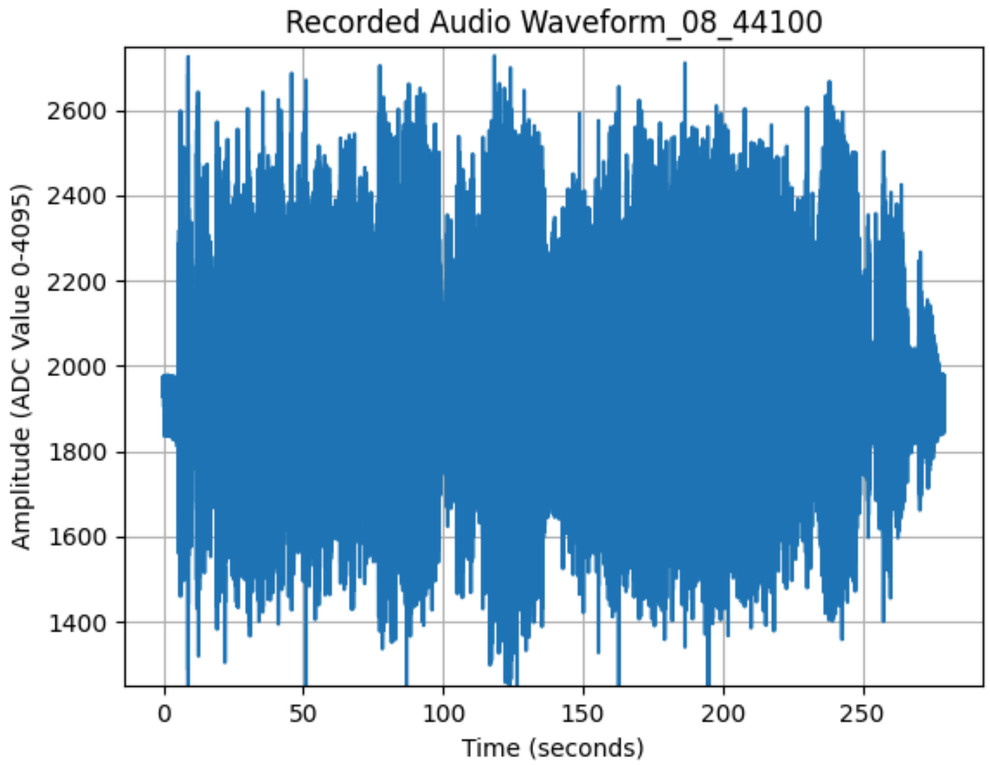
Python CLI

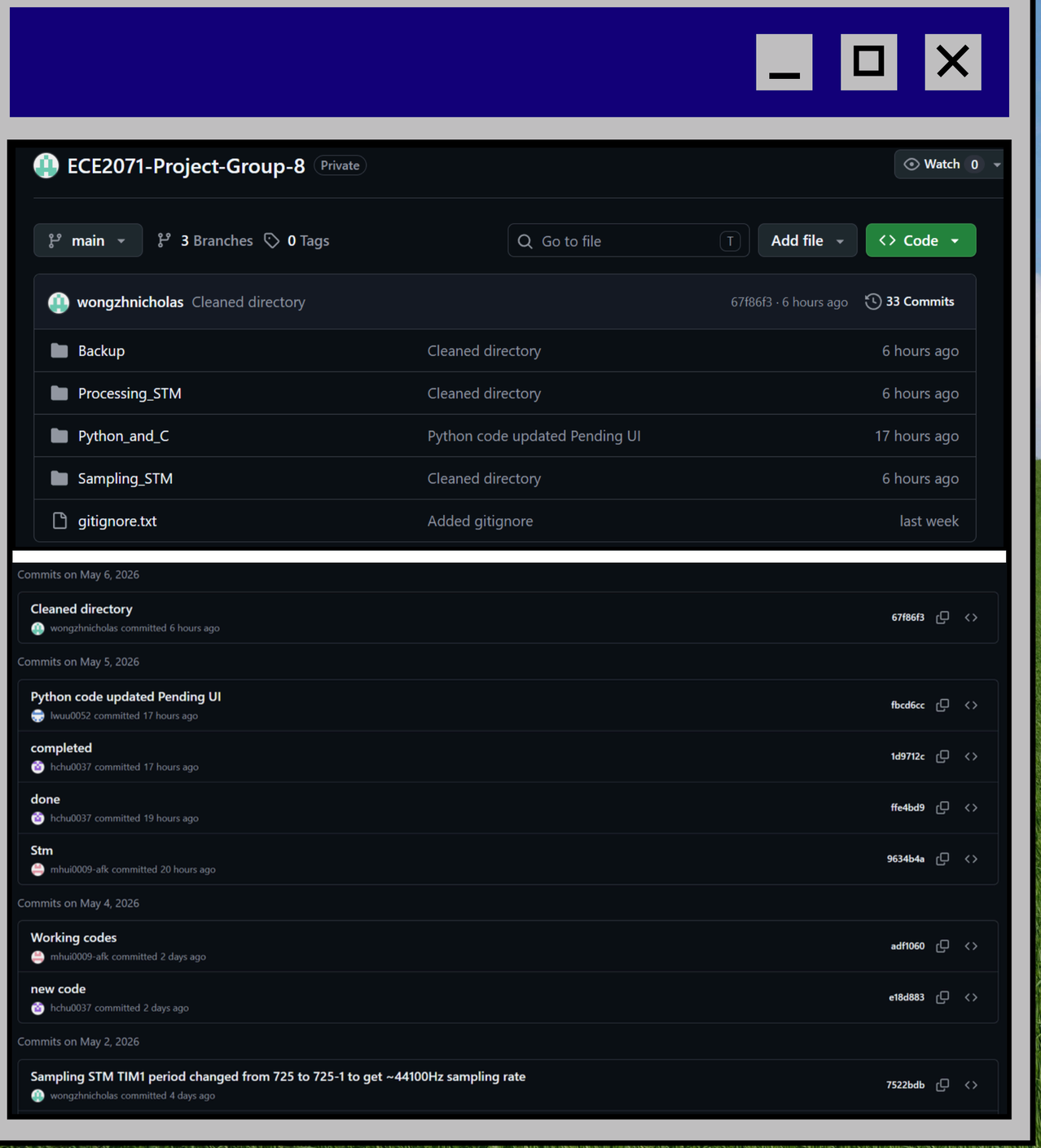
- **Serial Acquisition:** A Python script monitors the COM port to receive incoming USART data packets in real-time.
- **WAV Conversion:** Uses C-based program (called by Python) to perform conversion from binary file (.txt) to audio file (.wav).
- **Data Logging:** Formats the processed audio into standard file types (like .wav or .csv or .png) for long-term storage.
- **User Control:** Provides a Command Line Interface (CLI) for the user to view system status, select mode: distance trigger or manual recordings, and set proximity thresholds.

Results Obtained

	A	B
1	Sample Rate	44100
2	1968	
3	1967	
4	1966	
5	1967	
6	1967	
7	1967	
8	1968	
9	1968	
10	1968	
11	1968	
12	1968	
13	1969	
14	1970	
15	1970	
16	1970	
17	1969	
18	1969	
19	1969	
20	1969	
21	1968	
22	1967	
23	1966	
24	1965	
25	1966	
26	1966	
27	1967	

1048552	2050
1048553	2042
1048554	2034
1048555	2026
1048556	2018
1048557	2011
1048558	2005
1048559	1997
1048560	1989
1048561	1981
1048562	1970
1048563	1962
1048564	1955
1048565	1948
1048566	1942
1048567	1936
1048568	1930
1048569	1922
1048570	1916
1048571	1909
1048572	1906
1048573	1901
1048574	1897
1048575	1894
1048576	1889





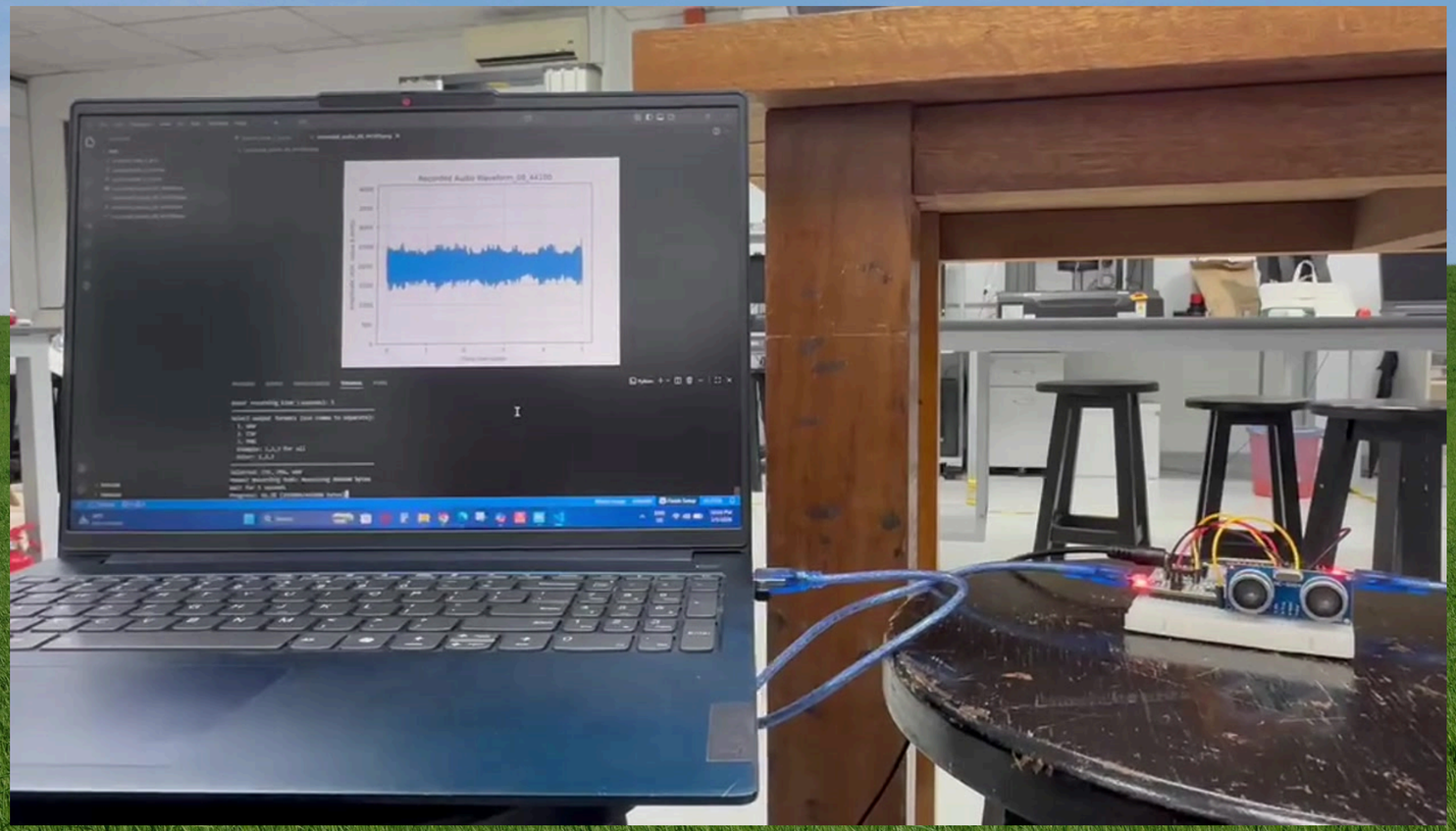
Project Management

Github was use so that every member is able to share and track one another's code at the same time.

It helps members to gauge the current progress of the project and see if any fixes are needed to the current version of the codes.

Short Demo

Short video to show how our system works



Conclusion

In the end, our group was able to implement the system successfully, with all of the improvements. It has notable functions such as:

- @44.1kbps @12-bit ADC data sampling and transmission throughout
- Pass it to the computer to convert it into a png, csv and wav file.
- Works in both manual & distance trigger mode
- Moving average filter & outlier rejection implemented

Thank you!